

Sanjana Kiran Urs

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SUMMARY

AI/ML Engineer with graduate training in Computer Science (AI/ML), focusing on Software Engineer and Data Scientist. Strong foundation in mathematics, statistics, and data science, with hands-on experience in regression modelling, machine learning, NLP and cloud technologies. Proficient in Python, SQL, and R, with experience analysing large datasets, data warehouse ETL, building reporting pipelines, and participating in model and design reviews to improve system quality. Motivated to learn through mentorship and apply analytical expertise to deliver data-driven solutions that enhance customer experience.

EDUCATION

University at Buffalo, Buffalo, New York

Master's in Computer Science, AI/ML Specialization

August 2024 – December 2025

GPA: 3.70/4

Coursework: Computer Vision and Image Processing, Statistical Data Mining II, Deep Learning, Data Models Query Language, Intro to Pattern Recognition, Operating Systems, Machine Learning, Algorithms, Data Intensive Computing, Computer Security

JSS Science and Technology University, Mysore, India

Bachelor of Engineering in Information Science

August 2019 – August 2023

GPA: 3.87/4

Coursework: Artificial Intelligence, Data Mining, Database Management System, Object Oriented Programming in C++, Software Engineering, Linear Algebra, Data Structures.

EXPERIENCE

Cedars Sinai Medical Center - Slomka Lab, Los Angeles, California | Visiting Graduate Student

July 2025 – August 2025

Cardiovascular outcome and disease prediction using AI | Pandas | NumPy | XGBoost | Python

- Developed an interpretable XGBoost model to detect coronary artery disease using PET-derived imaging biomarkers and clinical variables from 11 multicenter cohorts. Implemented LOSO cross-validation (train on 10 sites, test on 1) to ensure cross-site generalizability, improving AUC from 0.81 to 0.84.
- Coordinated integration and standardization of cardiac PET/CT imaging biomarkers across multiple sites, built quality-control pipelines and harmonization protocols, which reduced data ingestion time and ensured consistent analysis

ACADEMIC PROJECTS

Emotion Detection in Social Media Post using Sentiment Analysis | Pandas | PyTorch | NumPy | Scikit-learn | Python

- Fine-tuned a Transformer-based classifier on Google's GoEmotions dataset using strong analytical skills, implementing text cleaning, WordPiece tokenization, and class weighting to handle severe label imbalance.
- Architected and benchmarked three models - RoBERTa, RoBERTa+CNN+NRC-lexicon, and DeBERTa, applying critical thinking to achieve 85% accuracy, reducing test loss from 0.68 to 0.11, and maximizing F1 and ROC-AUC scores.
- Designed an end-to-end data pipeline for large-scale text ingestion, preprocessing, and feature transformation, incorporating structured data modelling of emotion labels and metadata to support efficient training, evaluation, and experimentation workflows.

Web Application for Basic Computer Vision Tasks | OpenCV | Python | HTML | JS | Flask

- Built an interactive web application that allows users to upload images and visually explore fundamental computer vision techniques such as image preprocessing, feature detection, camera calibration, and stereo image rectification.
- Designed a modular, end-to-end backend architecture with a highly collaborative approach, supporting seamless image processing and future extension to advanced vision tasks such as 3D reconstruction.

Identification and Classification of adherence in Cardiovascular Disease (CVD) patients using AI techniques

| Python | Dart | Flutter | MySQL | JavaScript

- Developed an Android application patient data collection while ensuring ethical handling of sensitive health data and data privacy considerations and identified if the patient's health status was healthy or not healthy
- Experimented by applying various machine learning algorithms such as Artificial Neural Network, Support Vector Machine, Random Forest to employ for training the model and obtained an accuracy of 88.75% for ANN

Vision Transformer based Cats vs Dogs Classifier | PyTorch | NumPy | Python | Hugging Face | Gradio

- Fine-tuned a pre-trained ViT model for cat vs. dog image classification using aggressive augmentation, mixed-precision training, and cosine-annealing LR scheduling, achieving 92.7% test accuracy
- Packaged the model into a Gradio web interface for interactive image upload and prediction

TECHNICAL SKILLS

Languages: Python, C, HTML, CSS, SQL, R

Libraries: NumPy, Pandas, SkLearn, Matplotlib, Scikit-learn, PyTorch, TensorFlow, OpenCV, Huggingface

Tools & Frameworks: MySQL, PostgreSQL, Tableau, Spark, Hadoop, Linux, Git, Docker, AWS, MS Excel, Power BI, Jira, FastAPI

Soft Skills: Problem-solving, Communication skills, High collaboration, Analytical skills, Critical thinking, Decision making

LLMs and GenAI: RAG, Transformers, Prompt Engineering